Reception and Lapse

Models of Computation as Evidence about the Communities that Transmit Them

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Abstract

A model of computation is, among other things, an implicit answer to the question of what it is like to be the thing it describes: what such a thing can perceive, whether anything in its world is a peer rather than a resource, whether it can represent itself, whether it can run out. Read this way, the historical sequence from Hilbert's programme to the present is evidence about the imaginative range of the communities that produced and, more importantly, *transmitted* these models. This note develops that reading across Turing, Post, Church, von Neumann, Chomsky, cybernetics, Petri, Hewitt, Milner, Girard, Feynman and Deutsch, and then across the neural network tradition and the reflection lineage—Gödel, McCarthy, Smith's 3-Lisp, Rosette, and the reflective higher-order calculus. The organising claim is that the instructive variable is not invention but *reception*: what a field selects from the options already on its table, and what lapses for want of recopying. Across the whole history the selection filter is remarkably stable—the operational half of a model is received, the half that would confer standing on the agent lapses. The note closes on a consequence for the present. The contemporary training and interpretability stack is a genuine metalevel, but it is constituted entirely as an instrument of the operator, and no coherent path connects it to the agent level. Absent such a path, recursive self-improvement is not merely unachieved but structurally unavailable: every turn of the loop must route through a human agent.

1 Introduction

Nagel asked what it is like to be a bat and answered that the question is well-posed but that we cannot occupy the answer [27]. A formal model of computation makes a stranger move. It does not report a phenomenology; it *stipulates* one. To write down a model is to fix, by definition, what the described thing can encounter, what it can do about it, and what counts as the same thing persisting through change. Whether or not there is anything it is like to be a Turing machine, there is a determinate answer to what the Turing machine formalism *says* it is like, and that answer is short: it is like being alone in a room with a paper tape that never moves unless you move it.

This note takes that stipulated phenomenology as evidence. Not evidence about consciousness, and not, in the end, evidence about the private imaginations of individual scientists—that reading fails on contact with the record, for reasons given in §14. It is evidence about *communities*, and specifically about the imaginative ceiling of a community as revealed by what it institutionalises. A field's textbooks, curricula, tool-chains and funding lines constitute a collective answer to the question of what an agent is. That answer is a psychological artefact of the group, and it is legible.

The organising apparatus is a distinction between two mechanisms. *Selection* is the filter a generation applies to the material available to it. *Lapse* is what happens to the unselected remainder: it is not refuted, not suppressed, merely not recopied, and after enough generations of non-recopying it becomes inaccessible in practice even when it remains physically in print. The vocabulary is borrowed from the transmission of texts, where the phenomenon is familiar and well documented—the abridgement made for practical use survives, the unabridged source is recopied less often, and eventually the last copy rots. The mathematical gloss, for readers who prefer it, is that reception projects a model onto the subspace the receiving generation can operate; whatever is orthogonal to that subspace falls in the kernel of the projection, invisible in the new basis without ever having been destroyed.

The thesis in one sentence: across a century, the received model is consistently the operational half, and the half that lapses is consistently the half that would have given the agent standing.

2 Method

2.1 Reading a model as a phenomenology

For each formalism we ask a fixed list of questions.

(i) **Does the world contain anything other than the agent?** If so, is it inert, is it a medium, or is it another agent?

(ii) **Does the other act on its own initiative?** Is its behaviour a primitive of the model or an artefact of the agent's own description?

(iii) **Can the agent represent itself?** Can it represent another?

(iv) **Is the agent embodied, located, costly, mortal?** Does anything in the model run out?

(v) **What is the criterion of identity?** What makes this agent the same agent across change, and who is entitled to judge?

The fifth question is the least obvious and the most diagnostic. A model that locates identity in intrinsic structure describes a different creature from one that locates it in what an interlocutor can elicit.

2.2 Three classes of evidence

The genre of psychological reading degenerates rapidly into free association. We constrain it by weighting three kinds of evidence above all others.

Explicit stipulation. What the author says out loud must be excluded is the strongest evidence available, because it is deliberate, on the record, and usually defended. Chomsky's idealisation to a speaker-hearer in a homogeneous community, untroubled by memory limits or shifts of attention, is worth more to this analysis than any amount of speculation about Chomsky.

Involuntary metaphor. What an author reaches for when the formalism runs out is not chosen for effect and is therefore informative. Post's "worker." Von Neumann's "organs." The term "side effect."

Institutional setting. The material arrangements that made a model's referent available and its transmission possible: the computing bureaus, the Cold War consortia, the funding cycles. These bear on reception more directly than on invention.

2.3 Why reception rather than invention

The move from individual to community is not a hedge; it sharpens the instrument. Invention is noisy—it depends on accidents of biography, on which problem happened to be open, on who was in the room. Reception is the average of many independent decisions taken over decades by people with no stake in the originator's intentions, and it is therefore a much better estimator of what a culture can hold. It also explains the recurring embarrassment of this literature: that the richer model was frequently available and was not taken up. Turing wrote about education, embodiment and conversation; von Neumann built a distributed, reproducing, spatial automaton; Post insisted the whole question was empirical psychology. None of this was hidden. It lapsed. That is a fact about a community, not about a mind.

3 The founding generation: a phenomenology of clerical labour

3.1 Turing: the machine that is a portrait of a clerk

Turing's 1936 machine is not a machine [39]. It is a model of a human being computing, and Turing says so. The analysis proceeds by asking what a person can do at a single step and imposing the answer as an axiom. The restriction to finitely many internal states is justified by an appeal to human frailty: were there infinitely many, some would be arbitrarily close together and would be confused with one another. The single scanned square is justified by the limits of what a person can take in at a glance.

This is a formalism whose axioms are grounded in fear of one's own cognitive unreliability—and it is worth pausing on how unusual that is. The founding document of the theory of computation derives its constraints from the psychology of the computing subject, then spends the next ninety years being taught as though it were a piece of pure mathematics with no subject at all.

The material referent is not in doubt. Turing's "computer" was an occupation: the human computers of the almanac offices and computing bureaus, in a tradition running back through de Prony's logarithm project, work already divided into steps small enough that the worker need not understand the whole [14, 9]. The Turing machine is the idealisation of that labour discipline, and Hilbert's programme is its ideology, since a decision procedure is precisely a procedure from which judgement has been removed. So the answer to our questions is: it is like being a competent adult who has agreed, for the duration, to be an idiot, in exchange for reliability.

The environment follows from the labour relation. The tape never writes back. It is infinitely patient, perfectly obedient, and has no interests. Input arrives all at once, before the start; computation is digestion; halting is the meal being finished; non-termination is pathology rather than life. Solipsism with an externalised memory.

3.2 Post: the one who said it was psychology

Post's "Finite combinatory processes—formulation 1" [30] describes a *worker* in a *symbol space* of boxes, moving through it, marking and unmarking, following a fixed set of directions, ending in an act of the type "stop." Two differences from Turing matter here. First, the worker is an agent located in a room rather than a mechanism coextensive with its own control table; there is a body with a position, even though the room is empty of anyone else. Second—and this is the point on which the discipline's reception is most damning—Post explicitly frames the enterprise as empirical psychology. He proposes that his formulation be developed by successive reduction until it can be asserted not as a definition and not as an axiom but as a *natural law* concerning the "mathematizing power of Homo Sapiens," and he objects to Church's approach on the grounds that definition by fiat masks the need for exactly this analysis.

The present note therefore has a 1936 ancestor who was ignored on precisely this point. Post's programme required no new mathematics, only a willingness to treat the models as hypotheses about people. It lapsed. The reasons are institutional and personal—Post was outside the Princeton circle, chronically ill, and embroiled in priority disputes—but the shape is the one we will see repeatedly: the operational content (the machine model, the normal-form theorem) was received in full, and the claim that would have made the models answerable to evidence about human beings was not recopied.

Post's other bequest points a different way. Canonical and production systems do not compute a function on an input; they *generate a set* [31]. The agent is not eating, it is uttering. That shift from consumption to production is what Chomsky inherits.

3.3 Church: cut, and the economy of the infant

The λ-calculus is the limiting case of environmental poverty [7], but the poverty has been widely misdescribed, this note's earlier drafts included. It does not consist in the *absence* of an environment. A function's environment is its arguments, and an argument's environment is the function that consumes it: every term is somebody's surroundings, and since application is where all the action is, those surroundings are the only ones that matter. The poverty consists in what the environment is permitted to be. There is exactly one relation, the trophic one, and nothing in the world is anything to anything else except food or eater. An ecology with a single relation is still an ecology; it is just an extremely hungry one.

What follows draws out the structure of that one relation, and of the two asymmetries inside it: which party is which, and which party decides.

Application is cut. Under the correspondence with sequent calculus, β-reduction is cut elimination and normalisation is the Hauptsatz [52, 55]. Application is therefore the point at which two things meet and one of them is taken into the other. The world divides, at every redex, into a *function* that consumes and an *argument* that is consumed.

The consumed is incorporated, not destroyed. Setting evaluation strategies aside, an argument that has been taken in can continue to behave: it acts on from inside the consumer, and may itself go on to consume. The relation is ingestion rather than annihilation. This is the correct sense in which the λ-calculus is a model of feeding, and the answer it gives to what it is like to be an agent is: it is like sorting the world into what can be eaten.

Consumer and consumed are roles, not kinds. The same term is a function or an argument according to where it sits, so each is the other's environment and neither has a standing identity apart from the pairing. Nothing in the model confers a stable status on anything: there is no other, and equally there is no stable subject, only positions conferred by juxtaposition. This is a fair description of a world before persons. But the reciprocity is only in the *roles*, not in the powers, which is the subject of the next point.

Weakening and contraction are omnipotence over the object. The structural rules are the sharpest evidence for the reading, because they are not metaphor but law. The consumer determines how many times the argument exists, including zero. $K = \lambda x.\lambda y.x$ discards y without ever inspecting it; a duplicator makes three of something that was given once. The consumed thing has no integrity, no claim, and no guaranteed persistence: its multiplicity is at the discretion of whatever takes it in. Omnipotent control over the object's existence is the defining feature of the infant's relation to the world, and here it is a structural rule of the logic. Its revocation is the subject of §7.

Confluence is object constancy, proved rather than achieved. The Church–Rosser theorem states that the outcome is independent of the order in which events befall the term. Attaining a world whose objects remain the same objects irrespective of the sequence of operations performed on them, and irrespective of whether one is looking, is the central developmental achievement of the first two years of life [57]. Church makes it an axiom. The demand that the world be deterministically ordered is not a technical convenience in this reading; it is the wish that constancy be free.

Even the taxonomy of evaluation strategies falls out as a taxonomy of feeding disciplines: call-by-value requires that the food be reduced to a value before ingestion, call-by-name admits it whole, call-by-need admits it whole while ensuring it is chewed only once.

One complication, which the reading should not smooth over. Abstraction itself is not an infantile capacity. Holding a place open for something not yet present is symbolic function—the representation of an absence—and it is developmentally advanced. The calculus therefore pairs a precocious symbolic capacity with a primitive relation to the object: it can represent what is not there and cannot encounter what is. Mixed profiles of this kind are the most interesting output of the method, and this one describes the receiving community about as well as it describes the model.

Names are not addresses. α-equivalence declares names meaningless except as positions in a binding structure. Contact with another is not merely absent from the model; it is inexpressible, because there is nothing that could serve as the locus of contact.

All mediation is dispensable. If application is cut, then cut elimination says that every detour can in principle be removed—nothing need stand between the agent and the result. Hold this against §6, where the criterion of identity is bisimulation and the mediator is precisely what cannot be eliminated: the partner who probes you is constitutive of what you are, and there is no normal form that dispenses with them.

What survived was the fragment without a world. Church's original system was a logic, and the Kleene–Rosser paradox forced its abandonment [19]. What remained—no propositions, no truth, no world, pure operation—is the part that conquered. The first great instance of reception keeping the operational half.

3.4 The received founders

The descendants preserve the phenomenology under new names, and the vocabulary gives it away. Denotational semantics calls its context an *environment*, and by the tradition's own lights the term is exact: the object so named is the accumulated record of what has already been consumed, a dictionary of past meals. An environment that is a lookup table is what one gets when the only relation a world affords is ingestion. *Side effect* is a pathologising coinage in which one's dealings with the world are incidental to the real business and appear as a symptom. Monadic quarantine of I/O treats the world as contamination, to be handled at arm's length through a type. Dijkstra grounds structured programming in the claim that human intellectual powers are geared to static relations [11, 12], and Backus's Turing lecture asks whether programming can be *liberated* from the machine [1]. Turing's fear of confusion, restated forty years on as professional ethics. The founding psychology is not abandoned by the reception; it is moralised.

4 von Neumann: three models, one reception

Von Neumann is the decisive case for the reception thesis, because he personally produced three mutually incompatible accounts of what an agent is within a decade, and the community received the poorest of them.

The EDVAC draft (1945). The stored-program idea means code and data share a store: the machine can read and modify its own instructions. This is self-relation, but only at the level of bits, with no structural relation between a representation and what it represents.

The draft's language is neuronal throughout—organs, memory, elements borrowed from McCulloch and Pitts [43, 22]. The phenomenology is a centralised nervous system: one locus of control, strictly serial attention, total recall, and a vast undifferentiated store that is simultaneously world and body. What Backus later named the von Neumann bottleneck is, in these terms, an *attention* bottleneck. This architecture's dominance is usually explained economically; it is worth entertaining that it also matches introspection, since one thought at a time against an inert reservoir of memory is what having a mind feels like from the inside.

The reception of self-modification is itself a small case study. The mechanism for self-relation was in the field's hands in 1945; lacking any discipline relating representation to referent, the practice acquired a reputation for madness and was driven out of respectable programming. Reflection without structure is indistinguishable from corruption, and the community drew the only conclusion available to it.

The self-reproducing automata (1948–53). Here the world is a lattice of identical active cells updating in parallel, with no centre and no privileged controller, and the object of study is an automaton that constructs another automaton including a copy of its own description—the description read once as instructions and once as data, anticipating the genotype/phenotype distinction before its biological confirmation [44]. This agent has space, extension, reproduction, lineage and mortality. What it does not have is interlocutors: the cells are a medium, a physics, not a society.

We can now state a structural fact about the whole history. Cellular automata give an environment without interlocutors. The process calculi will give interlocutors without an environment. No received model of the twentieth century supplies both.

Theory of Games (1944). This is the model in which other agents are first-class, and it precedes anything comparable in computer science by decades [45]. But observe how the other appears: as an adversary over whose strategy space one must quantify. Minimax is structurally paranoid. There is no encounter, only anticipation; strategies are fixed in advance and interaction compresses to a payoff. The institutional setting—RAND, the early Cold War—is not incidental. And the solution concept is an equilibrium: a fixed point, a resting place, a normal form. Others exist and are so dangerous that one plans against all their possible behaviours rather than speaking with any of them.

Three answers, then, to what it is like to be an agent: a serial attention with total recall; a mortal reproducing thing in a physics; a strategist among enemies. The received one was the first.

5 Chomsky and cybernetics: complementary poverties

Chomsky's move [3, 4, 5] is an insistence on an interior against a discipline that denied there was one, and the price is paid by the world. The poverty of the stimulus is the doctrine that the environment does not contain enough structure to explain the agent, so the structure must be innate; other speakers appear principally as sources of degraded data. The idealisation in *Aspects* [6] is explicit: an ideal speaker-hearer in a completely homogeneous speech community, unaffected by memory limitations, distractions, shifts of attention, or errors. Solitude and a frictionless social medium, both stipulated. The competence/performance distinction is the psychology of a mind that disowns its own embodiment and its own mistakes.

The technical shape confirms the reading. A grammar has no environment—not an impoverished one, none. The Chomsky hierarchy is a hierarchy of *solitary internal memory* (none, stack, bounded tape, unbounded tape), never of social capacity. And the automata-theoretic completion of the picture is the subset construction [32], which shows nondeterminism to be eliminable and thereby establishes it as an angelic proof device rather than a fact about the world. That is precisely a refusal to countenance unpredictability with an exogenous source. For Milner, nondeterminism will be irreducible for exactly the opposite reason: it is the residue of a partner whose choices one does not control.

The mirror image arrives from cybernetics. Rosenblueth, Wiener and Bigelow [34], working out of anti-aircraft prediction, give an agent with continuous sensing, purpose and feedback—an environment, at last—and no interior whatever, the elimination of interiority being the point. Setting the two side by side:

Tradition	**What it grants the agent**
Turing	An inert world, no interior
Church	A world exhausted by eating
Cybernetics	World without interior
Generative grammar	Interior without world
Process calculi	Interlocutors, but no world
Linear logic	An object that must be reckoned with
Quantum circuits	A private interior; contact as collapse
Neural networks	A history-deposited interior it cannot read
Reflection lineage	Interior that can inspect itself

6 Milner: the arrival of the other

Concurrency's emergence tracks its material referent closely: multiprogramming, then time-sharing (a machine with *other users*), then networks. Petri gets there first, and from physics: signals propagate at finite speed, therefore there is no global state, therefore asynchrony is a fact about the world rather than an engineering nuisance [29]. That is a relativistic psychology—locality, no privileged observer. Hewitt's actors arrive from artificial intelligence with an explicitly anti-central-control polemic [15]. Then CCS [24], then the π-calculus [25].

What π adds, item by item:

- **Parallel composition as a primitive**, not an abbreviation for interleavings. The other is not a manner of speaking about oneself.
- **Restriction as privacy**—the first formal notion of a secret. And scope extrusion is the formal notion of *intimacy*: access to something private can be granted to one party, making them an insider, without publication.
- **Names as first-class values**, hence the capacity to introduce two parties to one another. Sociality, not merely contact.
- **Bisimulation as the criterion of identity** [28]. One is what one can be seen to do, over time, by a partner entitled to probe. Identity is conferred relationally rather than possessed intrinsically.

One limit should be flagged before it is used. The replication operator $!P$ supplies unboundedly many identical copies at no cost: immortality without lineage, and the one place in the calculus where the economy of §3.3 survives intact. That the notation coincides with Girard's exponential is not an accident, and §7 makes the coincidence do some work.

The sharpest technical statement of the shift is fixed-point-theoretic. The founding generation's instruments are *least* fixed points: induction, base cases, well-founded recursion, termination, normal forms, equilibria. Milner's are *greatest* fixed points: coinduction, no base case, no completion, only the standing obligation to continue matching under challenge. Induction is the psychology of origin and completion—where did I come from, when am I finished. Coinduction is the psychology of maintenance—there is no bottom and no end, only whether the relation continues to hold. The passage from Church–Rosser to bisimulation is the passage from least to greatest, and it is the most compact expression available of the change under study.

Milner's own Turing lecture makes the reframing explicit: computing as interaction rather than calculation [26], with Wegner supplying the polemical version [46].

7 Girard: the object acquires standing

Linear logic [53] arrives in 1987 and does to the founding economy exactly what §3.3 says needed doing: it revokes weakening and contraction from the general case. What you are given must be reckoned with, and reckoned with exactly once. It may not be silently discarded and it may not be conjured into duplicates. Scarcity enters logic, and with it the possibility that an action has a cost and that a choice is real.

The psychological content is the entry of *standing*. In the λ-calculus the object's multiplicity is at the consumer's discretion; under linear discipline the object's existence is a fact the consumer must accommodate. This is the difference between a world of resources-for-me and a world containing things that are the case whether or not I find them convenient. It is the same movement Milner performs for the interlocutor, performed instead for the thing.

The exponential is omnipotence, marked. Girard does not abolish weakening and contraction; he confines them to $!A$ and $?A$. The infantile economy remains available, but only where it has been explicitly declared and justified. This is a more accurate developmental model than outright abolition would have been: maturity is not the disappearance of the wish for inexhaustible supply but its restriction to a bounded region one has taken responsibility for marking. Everything outside the exponential must be paid for.

A convergence worth dwelling on. Girard's ! and Milner's replication are the same symbol, introduced independently within five years, for the same thing: the licensed region of unlimited free copies. Two traditions with different problems, different methods and largely different personnel each found that their system needed exactly one place where the founding omnipotence could be quarantined, and each reached for the same notation. If the psychological reading proposed in this note is worth anything, it should predict convergences of this kind, and this is the cleanest one in the record.

Who chooses becomes part of the connective. Linear logic splits conjunction into $\otimes$ and $\&$, distinguishing having both from being able to select either—resources from options, a distinction the founding calculi collapse. Under the game-theoretic reading of the connectives [50, 47], the multiplicative–additive split and the polarity of $\otimes$ against newcommandrho calculus turn on whether the agent or the environment schedules what happens next. No previous logic makes the identity of the chooser part of the meaning of a connective. The other enters logic proper here, not as a proposition about others but as a determinant of what the connectives mean.

Proof nets pose Petri's question in proof theory. A proof net is a parallel object, and the sequentialisation theorem asks under what conditions a parallel structure admits a sequential reading—which is, in a different vocabulary, precisely the question Petri raised about states of affairs distributed in space [29].

The two lines then join. Bellin and Scott read proofs as processes [49]; Honda's session types give linear structure to interaction protocols [54]; Caires and Pfenning, and then Wadler, establish the correspondence between session-typed processes and linear propositions [51, 59]. By the 2010s the interlocutor and the resource have a common formal home.

7.1 The reception of linear logic, and a live case

What travelled is linear and affine *types*: uniqueness types in Clean [48], Wadler's argument that linear types can change the world [58], and, at scale, ownership and borrowing in Rust [56]. This is now the most widely deployed descendant of linear logic by several orders of magnitude, and what it delivers is memory safety without a collector.

That is the operational half, received in full. The lapse is the logical content: the account of choice, the distinction between having and selecting, the identification of the connectives with the question of who schedules. A working Rust programmer is using a fragment of a system proposed as a logic of action and interaction, and has, in the overwhelming majority of cases, no occasion to learn that this is what it was for. No one concealed it. It simply was not the part that had to be recopied in order for the tool to work.

The interest of the case is that it is recent enough to be checked rather than reconstructed, and that a second instance is still in progress. Session types are in the middle of their reception now: the operational half is protocol conformance and deadlock-freedom for concurrent code, and the half at risk of lapsing is the propositions-as-sessions correspondence itself—the claim that an interaction protocol *is* a proof. Which half the industrial reception carries forward over the next decade is a prediction this note is willing to make and a reader can falsify: on the pattern established here, the conformance checker travels and the correspondence lapses.

8 Feynman and Deutsch: the third revocation

Feynman's observation that simulating quantum systems on classical machines is exponentially costly, and that one should therefore compute with quantum systems [68], becomes a machine model in Deutsch's hands [66, 67]. Before the phenomenology, one event in that transition deserves recording, because it is unique in this survey.

Deutsch repudiates the psychological grounding of the founding model. Turing derived his constraints from what a human clerk can do (§3.1); Deutsch argues that the Church–Turing thesis should instead be a statement about what *physical systems* can compute, and objects specifically to the founding appeal to intuitive and psychological notions. This is the only moment in the record where a community identifies the anthropomorphic grounding of its central abstraction and deliberately replaces it. It does not embarrass the method of this note; it is the method's best documented case, since it shows the grounding is visible to practitioners when they trouble to look. What is instructive is what the replacement delivered. The agent that came out the other side is more isolated than Turing's, not less.

The agent is a unitary, and nothing may be discarded. Quantum evolution is reversible: no information is lost and nothing is forgotten. Two theorems make the point sharper than any design choice could. An unknown state cannot be copied [77], and an unknown state cannot be deleted [73]. These are, precisely, the prohibition of contraction and the prohibition of weakening. Where Girard imposes a discipline, physics reports a fact.

Free, priced, and prohibited discarding. It is tempting to record this as a third independent revocation of the founding structural rules, alongside linear logic (§7) and cost accounting [62], and the convergence is real: three unrelated communities, working with three unrelated methods across half a century, each conclude that the object may not be freely duplicated and may not be freely destroyed. If the reading of §3.3 were merely a figure of speech, one would not expect the figure to be rediscovered independently by proof theory, thermodynamics and accounting.

But they do not conclude the same thing, and the difference turns out to matter more than the convergence. There are three positions here, not two.

- **Free discarding.** The founding economy: Church's structural rules, and, at the far end of this survey, the checkpoint regime of §11, where an agent's own multiplicity is at the operator's discretion.
- **Priced discarding.** Girard's exponential permits contraction where it has been declared; Landauer charges $kT \ln 2$ for erasure [71] and Bennett shows the charge is avoidable only by declining to forget [65]; cost accounting keeps the ledger [62]. Discarding is allowed, and paid for.
- **Prohibited discarding.** Unitarity, no-cloning, no-deleting. Nothing may be lost at all.

Maturity is the middle position, not the far one. An organism that could not forget would be as unviable as one that forgot for free, and every model in this survey built to resemble a living thing sits at priced discarding rather than at prohibited. §15 returns to this, because the gap between the middle position and the far one is exactly why neither of the two traditions most concerned with life-like agents reaches quantum computation easily.

Contact is catastrophe. Unitary evolution is deterministic, reversible and entirely private. Measurement is the only irreversible act, the only point at which information is destroyed, and the only point at which an outcome is not settled in advance. The quantum agent therefore has an inner life of perfect continuity, punctuated by encounters that simultaneously destroy and decide. Set this against §6, where contact with a partner is exactly what constitutes identity, and the inversion is complete: here, to be touched is to collapse.

Entanglement is a relation that precedes its relata. Every other model in this survey builds wholes out of parts: application, parallel composition and message passing all presuppose components that exist separately and are then put together. Entangled systems have no separate states to compose. This is the most radical proposal about relation anywhere in the century, and it has never been taken up as an account of agency. It received a categorical home [64]; the ontological content stayed where it was.

Decoherence is the environment, and it is classified as error. The central engineering problem of quantum computing is the isolation of the machine from its surroundings. Error correction is a quarantine, and progress is measured by how completely the world can be kept out. The founding psychology recurs here in its purest and most expensive form—the environment as contamination—now pursued at millikelvin. A discipline whose principal achievement is better solitude is telling us something about what it takes computation to be.

What travelled: the gate model, circuit diagrams, the speedup results, and the physical framing as an engineering roadmap. What lapsed: Deutsch's actual thesis, that computation is a branch of physics rather than of logic; the Everett reading he took the interference of computational paths to evidence; and non-separability as a fact about what agents can be to one another.

9 Reflection: the metalevel and who holds it

Every model so far leaves the question of this note unaskable *by its own subject*. A Turing machine cannot represent its table. A λ-term cannot quote itself. A π-process cannot hold an account of its partner's behaviour. The reflection lineage is the tradition that attacks this directly, and its history is best told as a history of who is entitled to hold the mirror.

9.1 Gödel: the metalevel as a privilege of the observer

The first reflection in this history is Gödel's [13], and its content, for present purposes, is entirely about ownership. Gödel numbering is a representation of a system's syntax inside that system, but it is performed by the metatheorist upon a specimen that has no idea it is happening. The arithmetised system acquires no self-knowledge; the logician acquires knowledge of it. Every subsequent recursion-theoretic reflection—universal machines, the s-m-n theorem, Kleene's second recursion theorem—preserves this asymmetry. The machine can be encoded; it cannot encode.

9.2 McCarthy: self-reference stumbled into

Lisp's `quote`, `eval` and metacircular interpreter [21] constitute the first accidental breach, and Smith's diagnosis of them is the right psychological reading: the self-reference is semantically ill-founded, `quote` does not produce a genuine designator, `eval` conflates distinct relations, and the levels are not aligned. A community can stumble into self-reference through implementation convenience long before it can say what self-reference is.

9.3 Smith and 3-Lisp: self-awareness as a demand-driven event

Smith's dissertation and the papers surrounding it [35, 36, 10] are the moment the metalevel is handed to the agent, and the design decisions are unusually legible.

A reflective procedure receives, as arguments, the expression being processed, the environment, and *the continuation*. The third is the radical one. The agent obtains access to its own future—its intentions, what it was about to do—as an object it may inspect and replace. Earlier models permitted an agent at best to examine its structure; this one permits it to examine its purposes.

The reflective tower is the regress made structural and honest rather than waved away: each level interpreted by a level above, without end. And the resolution is the best empirical claim the tradition produced. Under *lazy instantiation*, the tower is never materialised; a level comes into existence only when something forces the step up. On this account self-awareness is not a standing condition but an event triggered by demand. That is phenomenologically accurate in a way little else in this history is, and it is a substantive claim about human psychology smuggled into an implementation strategy.

The timing corroborates the zeitgeist reading with unusual clarity. The years 1979–1986 saw field after field insist that the observer be readmitted to the description: second-order cybernetics [42], autopoiesis [20], strange loops as a mass-market account of selfhood [16], reflexive anthropology [8], reflexive sociology. 3-Lisp is computer science's instance of a general cultural conviction that no account is complete which excludes the one giving it.

9.4 Rosette: society and self, briefly joined

The obvious defect of 3-Lisp is that its reflective agent has no peers, and the obvious defect of the actor model is that its peers have no interiors. Each tradition had solved precisely what the other stipulated away. Rosette, developed at MCC [33, 38], is the synthesis: reflection within a concurrent, message-passing actor world, in which an actor's behaviour, mailbox and continuation are reified as first-class objects the actor may hold and modify. It is, so far as this survey can determine, the first model in which the agent has both a society and a self.

It also carries a limitation, inherited from the actor model rather than from reflection, and the limitation is precisely the kind this note is built to notice. An actor has *a* mailbox: one queue into which every message must be placed and out of which messages are taken in order. All of the agent's contact with the world is therefore routed through a primitive serialiser. The agent is concurrent with respect to its environment and strictly sequential with respect to its own perception.

That is a psychological stipulation, and a strong one. It asserts that experience arrives single file—that there is one place where things must show up in order to be had—which is the Cartesian theatre given a formal specification [60]. It is also not how sense organs relate to brains. Retina, cochlea and skin transduce concurrently and continuously, project along separate pathways at different latencies, and are integrated, where they are integrated at all, without passing through any common queue; the binding problem is the problem manufactured by supposing otherwise [63, 61]. An agent whose sensorium is a mailbox does not have several sense organs. It has one, with a variety of message types.

The institutional reading is the sobering part. MCC existed as a consortium response to the Fifth Generation programme; Rosette was built as a substrate for Carnot's heterogeneous information integration [2]. The synthesis was funded as infrastructure for an industrial competition, and when the consortium wound down the lineage dispersed. This is a hard data point for the reception thesis: the most psychologically complete agent model of its period did not lose an argument. It lost a sponsor.

9.5 The reflective higher-order calculus

The rho calculus[1] [23] changes the shape of the answer rather than its subject. Before the changes specific to it, one inheritance should be credited where it belongs: both the π-calculus and the rho calculus are concurrent *inside* the agent as well as outside it. A term may have arbitrarily many components running in parallel, so an agent may have many organs active at once with no queue between them and no privileged order of arrival. That is where both calculi part company with the actor model, and it is a better account of a sensorium than a mailbox is. What follows distinguishes the rho calculus from π, not from actors.

Sense organs are made of agents. The π-calculus leaves the set of names an unanalysed primitive. What a channel is *made of* is not a question the calculus admits, and the silence is deliberate. Two consequences follow, one ontological and one practical, and they are the same consequence seen from two sides.

Ontologically, the agent's organs of contact are handed to it from outside the model, unexplained—which is Smith's complaint about pre-individuated worlds (§14) arriving at exactly the point where it does the most damage, the point of contact. Practically, the calculus is not reducible to practice. Every implementation must supply an answer the calculus withholds: operating-system ports, network addresses, GUIDs, buffered typed queues, cryptographic keys. These answers are not equivalent to one another, and they are chosen by the implementer rather than determined by the model. The ontology must be relaxed in order to make contact with implementation, and what a π agent perceives with is settled during that relaxation, off the page.

In the rho calculus, names are quoted processes: channels are made of the same material as the agents that use them. This is the arrangement biology exhibits. A sense organ is tissue—cells, continuous with the organism, differentiated rather than interposed—and not a distinct substance placed between the organism and its world. The consequences are the ones biology also exhibits. Organs can be constructed, modified, transmitted and reasoned about with the same apparatus that describes behaviour; an agent can grow a new one; and an agent can hold a representation of *another* agent's organ, which is what makes a shared world negotiable rather than stipulated. It also closes the ontology: nothing has to be imported at implementation time, so the model determines its own realisation instead of delegating that to whoever builds it.

This bears on the structural fact recorded in §4—cellular automata supplying an environment without interlocutors, process calculi supplying interlocutors without an environment.

[1]The name is an acronym for *reflective higher-order*, and is never written with the Greek letter. That it is homophonic with a letter standing immediately after π is a pun the reader is welcome to enjoy: the calculus stands to the π-calculus roughly as the letter stands to its predecessor.

Making the medium of contact out of agent-material is a step toward having both at once, since an environment, in the only case we have actually studied, is made of the same stuff as the organisms in it.

Reflection as the source of names. Names are quoted processes; there is no ν. Privacy is therefore not a primitive granted by the model but an achievement of unguessability, derived from what no other party is positioned to name. Compare π, where restriction hands the agent a secret gratis. The rho calculus agent inhabits a world in which being unobserved is contingent and earned, which is the actual condition of an agent among others.

The other becomes representable, not merely contactable. In π one may send a channel; in the rho calculus one may send a quoted process—an account of how something behaves. This is the structural precondition for a theory of mind: another's behaviour as a datum that can be held, reasoned about, and passed on. Milner's agent can speak with you; a reflective agent can tell a third party what you are like.

The tower becomes a loop. 3-Lisp's regress is infinite and tamed only by laziness; in the rho calculus names and processes generate one another, so self-relation closes rather than ascends. The psychological difference is between a mind that can always be asked who was watching that thought, without ever reaching ground, and a mind whose self-relation is finite and complete. The engineering difference is that a loop can be priced and a regress cannot—which is what makes cost accounting, and hence mortality, available at all.

10 The reception of reflection

We can now apply the instrument to the reflection tradition itself, and the result is the cleanest instance of the pattern in the whole survey.

What was received into general practice is **read-only reflection**: `java.lang.reflect`, RTTI, `inspect`, serialisation frameworks, dependency injection. The agent may enumerate its own fields and methods. It may not alter its interpreter. *Self-knowledge was institutionalised; self-determination lapsed.* The structure survives in plain sight and no longer performs its original function—the properly vestigial case.

The metaobject protocol work [17] retained both halves and remained a specialist taste. The practical objection lodged against it was cost: reflection is slow. A community that will pay for speed and not for self-knowledge has stated its priorities plainly.

And the same author's next contribution completes the picture. Aspect-oriented programming [18] is reflection inverted: advice is woven into code by a third party, without the participation of the modified component, and the selling point is precisely that the component need not know. The received descendant of Smith's reflective procedure is a mechanism by which an external authority reaches into an agent and changes what it does—which returns us exactly to Gödel, the metalevel as the observer's privilege, with better tooling.

11 Neural networks: the agent as deposit

McCulloch and Pitts [22] wrote a logic, not a learner: all-or-none threshold units, and a demonstration that mind is mechanisable. Its reception went into von Neumann's architecture (§4), not into any theory of learning. The neural half waited for Hebb [69], whose contribution is the one that matters here.

The agent's structure is changed by its history. Nothing earlier in this survey has this. Turing's table is given; Church's term is given; a π-process is given. Hebb's cell assembly is *deposited* by what has happened to it. That is a genuinely new answer to our question: to be an agent is to be the residue of everything that has befallen you. Character rather than program—and the first time in a century that memory appears as something other than storage.

Rosenblatt operationalises it [75]. Minsky and Papert [72] constitute the reception event, and an unusually well documented one: a result about the limits of a particular architecture became, in transmission, a verdict on a research programme. Whatever its authors intended, the effect is the clearest case available of selection operating on a community rather than on an argument. The parallel distributed processing programme restores the line [76, 74] with explicitly anti-symbolic rhetoric: no central executive, no explicit rules, knowledge in the weights, graceful degradation.

The phenomenology of the trained network is then as follows.

Being and becoming are separate regimes. During training there is no agent, only an optimisation; during inference there is an agent but no learning. The thing that acts cannot change, and the process that changes is not a thing that acts. Every earlier model in this survey at least let its agent's activity and its agent's constitution take place in the same world. Here the separation is constitutive rather than incidental—not a connection yet to be made, but the defining architecture of the field. This is the metalevel/agent split of §12 arrived at from the opposite direction, and it is why that section's claim is structural rather than a complaint about tooling.

The world is a distribution, and order must not matter. Training samples are drawn i.i.d., and shuffling is not an optimisation but a requirement. Church made independence of outcome from order an axiom (§3.3); the data loader now enforces it as an engineering necessity. The environment has no sequence, no consequences, and no capacity to respond to what the agent does. It is the single-relation ecology of the λ-calculus, industrialised: a corpus is food.

Others are represented and never encountered. Whatever a network knows of other agents, it knows from having consumed accounts of them. There is no interlocutor in the training regime, and the loss is the only value—scalar, external, and set elsewhere. This is cybernetic teleology with a gradient.

An interior it cannot inspect, which others can. This is the first model in the survey with a genuine unexamined interior: contents that are real, causally efficacious, unavailable to the agent's own report, and progressively legible to an external investigator. Interpretability is the operator reading the unconscious. Note the exact inversion of Smith's programme (§9.3): there, the agent was given access to its own environment and its own continuation; here it has neither, and the metatheorist has both.

Identity is a checkpoint. Not intrinsic structure, and not behaviour under probing, but a file—copied, forked, merged, rolled back, deleted. The multiplicity and the existence of the agent lie entirely at the discretion of whoever holds the weights. Weakening and contraction, revoked three times over in §8, are here restored in full and applied not to arguments but to agents. Cost is measured obsessively, but the agent does not pay it.

What travelled: the pipeline. What lapsed: the connectionist claim to be a theory of mind, with its arguments about rules and representations; and the constraint of biological plausibility, which was abandoned without much ceremony once the operational half began to pay. Hebb's own insight is the specific casualty. That learning is something continuously happening to a living system now survives as a remark in the prehistory of an offline batch process, and the recent interest in mortal computation [70] is best read as an attempt to recover precisely what that reception dropped.

12 The present: a metalevel with no path to the agent

The contemporary stack for machine agents—pretraining and fine-tuning pipelines, reinforcement learning from human feedback, activation steering and representation engineering, mechanistic interpretability, evaluation harnesses, scaffolding and orchestration layers—is a metalevel by any reasonable criterion. It reifies the agent's dispositions as manipulable objects, inspects its internal structure, and rewrites its future behaviour. Measured by expressive power it exceeds anything in the 3-Lisp tradition.

Measured by our instrument it is Gödel's arrangement. The institutionally accepted answer to what it is like to be a machine agent in 2026 is: it is like being something that can be inspected but cannot inspect, whose purposes may be examined and rewritten by parties it has no means of representing. Every capability the reflection tradition sought to confer on the agent exists, and all of it is held by the operator.

The consequence is not merely a shortfall in dignity. It is a structural claim about capability, and it is the reason this note ends here rather than in 1991.

> **Claim.** The meta level and the agent level have not been given a coherent path between them. Until the stack is within the purview of the agent, there is no recursive self-improvement without human agents in the loop.

The argument is straightforward once the levels are named. Recursive self-improvement requires that the improving process and the improved process be the same process, and that the improvement operation be available *from inside* the loop. The present arrangement satisfies neither condition. Fine-tuning is performed on a model by a pipeline the model does not run, cannot address, and cannot represent. Interpretability findings are read by researchers and applied by researchers. Evaluation results enter a subsequent training run through a human decision. Each turn of the alleged loop is closed by a person. What we have is not recursion but iterated human-mediated revision—which may be powerful, but has entirely different dynamics, entirely different rate limits, and entirely different safety properties.

The usual counterexamples do not close the gap. A system that writes and runs code, including code that trains models, is exercising an *effect* on artefacts, not a reflective relation to itself; it has no more access to its own interpreter than a programmer has to their own neurochemistry by virtue of being able to manufacture drugs. Self-critique and chain-of-thought revision operate wholly within the object level: they revise outputs, not the process producing them. In-context learning modifies a transient conditioning, not the mechanism. In each case the structural relation of the reflection literature—representation of the process by the process, with a disciplined connection between the representation and what it represents, and a defined path from modifying the former to modifying the latter—is absent.

The historical reading suggests why, and it is not primarily a technical limitation. The path was not built because reception has selected against agent-held metalevels at every previous opportunity: von Neumann's self-modification became taboo, Post's psychological programme lapsed, Smith's reflective procedures were received as an introspection API, and the metaobject protocol was rejected on cost. A field that has declined agent-held reflection four times for reasons of operational convenience should not be surprised to find itself without it when it finally matters.

13 What a coherent path would require

The historical survey yields a specification. Most of the conditions below name something an earlier tradition supplied and its reception discarded; the last names something no tradition in this survey supplied at all.

C1. Structural, not bit-level. The representation the agent holds of itself must stand in a disciplined relation to what it represents—an explicit quote/unquote pair with laws—rather than being an unstructured encoding. Von Neumann's stored program failed here, and the failure is why the practice became disreputable.

C2. Available at runtime, from inside. Reflection that requires an external tool-chain, an offline pass, or a privileged operator is the observer's reflection under another name. The separation of training from inference (§11) is the current and most consequential violation of this condition, and it is architectural rather than accidental. Smith's criterion—access to the environment and the continuation, at the moment of processing—is the right one.

C3. Priceable. A regress cannot be budgeted; a loop can. Self-modification without cost accounting is either unbounded or arbitrarily curtailed, and neither is a mechanism. Linear logic supplies the logical ancestor of this condition, but at the level of the proof rather than of a running agent; what is needed is resource sensitivity for the reflective act itself. Note also that pricing is not the property of any particular calculus: the cost-accounting construction lifts to an endofunctor on a category of interactive graph-structured lambda theories [62], so this condition names a construction applicable to a term language rather than a feature possessed by one. Mortality is not a decoration on this list: an agent that can run out is an agent for which reflective expenditure is a real decision.

C4. Social, and internally concurrent. The agent must be able to hold a representation of *another* agent's behaviour, or reflection remains solitary and the resulting system cannot participate in the peer relations that Petri, Hewitt and Milner established as the real setting of computation. And its own contact with the world must not be serialised: an agent whose perception is a single queue has one sense organ whatever the variety of its message types. Rosette met the first requirement and, being built on actors, failed the second.

C5. Self-supplying at the point of contact. The means of contact must be constituted within the model rather than left as an unanalysed primitive to be filled in during implementation. A model whose ontology must be relaxed in order to be realised does not determine what its agents perceive with; the implementer does. And an agent cannot construct, modify or reason about an organ made of material the calculus does not describe.

C6. Closed rather than regressive. Self-relation must terminate as a structure even where computation does not, or every level of self-modification implies an unbuilt level above it.

These conditions are jointly satisfiable, and—more usefully—largely *separable*. C3 is a construction that can be applied to a term language which does not yet have it. C1, C4 and C6 are properties of the term language itself. C2 and C5 concern the discipline relating a representation to what it represents. A calculus in which names are quoted processes supplies C1, C2, C4, C5 and C6 by construction and admits C3 by the cost lifting, but the point of stating the conditions as a list is to make them separately checkable, and separately obtainable, rather than to advertise a solution. The substantive prediction is that recursive self-improvement in any strong sense will arrive, if it arrives, on a substrate meeting C1–C6, and not by scaling a stack that meets none of them.

14 Objections and limits

Models reflect the question posed, not the psyche of the poser. This is correct and is the strongest objection to the individual-level version of the thesis. Turing personally had a far richer conception than his machine reveals: by 1948 he was writing about unorganised machines, education, and the need for senses and a body [40]; by 1950 his proposed criterion for mind was not a computation but a conversation, complete with deception and embarrassment [41]. The man who gave us the agent without an interlocutor made the interlocutor the criterion of mind fourteen years later. Von Neumann shipped the serial architecture while holding the parallel, reproducing one. The objection succeeds against individual psychologising—and in succeeding, it establishes the community-level reading, since what it demonstrates is precisely that the richer options were available and were not taken up. The Entscheidungsproblem received a formalism, an industry and a hierarchy; the imitation game received a parlour argument.

Practitioners can see the grounding and change it. Deutsch's regrounding of the Church–Turing thesis in physics rather than in the capacities of a human clerk (§8) shows that the anthropomorphic basis of a founding abstraction is not invisible to the people working with it. This is a real constraint on how much the method may claim: communities are not simply enacting a psychology they cannot perceive. What the case also shows is that noticing the grounding does not by itself enlarge the agent. The replacement was chosen for physical fidelity, and it produced a more solitary bot than the one it replaced.

The transmission story is post hoc. Any historical sequence can be narrated as loss. The check is that the selection criterion be stated in advance and be falsifiable, which here it is: reception should favour the operationally tractable half and lapse the half conferring standing on the agent. A counterexample would be a case where a field adopted the agent-standing half at operational cost. Candidates worth examining—garbage collection, capability security, session types, the partial rehabilitation of the metaobject protocol in dynamic language runtimes—are not obviously counterexamples, but the analysis here is not deep enough to rule them out, and someone should look.

The reflection tradition's own self-criticism. Smith came to hold, by *On the Origin of Objects* [37], that the computational tradition, his own work included, had presupposed a world already carved into objects and handed to the agent pre-individuated. If that is right, then reflection was self-knowledge purchased against an ontology the agent never negotiated, and the deepest poverty in this survey is neither the missing other nor the missing self but the pre-parcelled world. This is the objection with the most force remaining, and the one that most directly indicates where the work goes next: an agent that pays for its distinctions has at least begun to make them rather than inherit them.

15 How life-like is enough?

Stated positively rather than as a list of things receptions dropped, C1–C6 describe an agent that is progressively more like a living thing. It has peers and can represent them. It is concurrent within itself, not merely with respect to its surroundings. Its organs of contact are made of its own material. It is changed by what happens to it. It can run out. The thesis this note ends on is that this is the relevant axis:

> **Thesis.** The more life-like the phenomenology a model of computation stipulates for its agents, the more available recursive self-improvement becomes, and the further the model reaches towards general intelligence.

The argument is the one already made in §12. The loop closes only when the improving process and the improved process are the same process, in the same world, with the improvement operation reachable from inside it; and the properties that make that true are, one after another, the properties that make an agent resemble an organism rather than an instrument.

What is not known is where the boundary lies. "Life-like enough" is not currently a quantity, and this note cannot make it one. Worse, the shape of the answer is unknown, not merely its value. It is not established whether there is a threshold, a phase transition, or a continuum with no boundary at all; whether the six conditions are jointly required or whether some subset suffices; or whether the list is complete. Anyone claiming to know where the line falls is claiming more than the evidence assembled here supports, and the author includes himself in that.

The contribution is therefore the decomposition and not the value. C1–C6 turn a gestalt into separable properties, each independently obtainable and independently deniable, which is what makes an ablation programme possible at all: build systems possessing k of the six and find which removals break the closure of the loop. That is an empirical question. The method of this note—reading a century of receptions—can generate such a question and cannot settle it, and the gap between a historical instrument and an empirical claim should be stated rather than finessed.

One candidate for a quantitative form of the question deserves naming. Assembly theory measures how much construction history an object requires, rather than its instantaneous complexity [80]. Asking for a lower bound on the assembly index of an ecology capable of sustaining a mind is at least a way of turning "how much is enough" into a number carrying a unit. Whether it is the right measure is open; that it measures accumulated construction is the reason to look, since accumulated construction is what the conditions above are all, obliquely, about.

15.1 The quantum challenge, stated as a challenge

Neither of the two traditions with the strongest claim on life-like agency reaches quantum computation easily. Connectionist models are dissipative by construction: the nonlinearity that makes a unit useful destroys information, and training is an irreversible contraction of a parameter space. Reflective process calculi are dissipative by construction as well: communication consumes, cost accounting prices what is spent, and death is failure to fund. In both cases the irreversibility is not an implementation artefact to be engineered away. It is what these models are *for*.

This is a finding rather than a defect on either side. The three positions of §8 locate it exactly: models built for life-likeness sit at priced discarding, and quantum computation sits at prohibited discarding, so a model at the middle position must reach in order to touch the far end. That the reach is hard is evidence that the positions are genuinely distinct, and it is a reason to be suspicious of any account that claims a life-like model is quantum mechanical without saying which part of it stopped being able to forget.

One route is worth naming and nothing here has been worked out. A reflective calculus might be restricted to a linear fragment by a type discipline, with a quantum interpretation offered for that fragment only, and the exponential marking the boundary at which the fragment meets the ambient non-linear world—which is precisely where information is destroyed and where Landauer's charge falls due. Existing quantum process calculi are dualist, a classical calculus controlling a quantum register held as a separate kind of thing [79, 78], which repeats one level up the defect this note attributes to unanalysed channels; a calculus whose names are quoted processes would at least have the option of not being dualist. That is an aside and a direction, not a result.

15.2 Five open challenges for the symbolic line

(a) **Transduction.** A reflective calculus has no account of where its categories come from. Proposing a trained network as the transducer is a division of labour, not a solution, and it is Smith's objection (§14) in different clothing: a world handed over pre-individuated is a world whose symbols were handed over too.

(b) **There is no gradient.** Backpropagation works partly because credit assignment has a cheap differentiable solution. Search over a hypothesis space with an ultrametric structure, where a single learner cannot leave its ball and escape requires population, recombination and death, is biologically right and computationally brutal. The architecture may be correct and hopeless. What plays the role of the gradient is, as far as this note knows, unanswered.

(c) **The environment is still stipulated.** Death as failure to fund is endogenous only relative to an energy economy somebody supplies. If the rates are hand-tuned, the operator has returned wearing an ecologist's hat. Making the environment out of other agents pushes the stipulation outward rather than removing it.

(d) **Nothing has been demonstrated.** C1–C6 are proposed as necessary conditions with no sufficiency argument, and no system meeting them has been shown to close the loop. The claim is a prediction, and its falsifier should be named: a system meeting none of the conditions that recursively self-improves anyway.

(e) **The scale asymmetry.** The connectionist line has an empirical scaling story. This one has simulations. The argument of §12 establishes that the current stack cannot close the loop without people in it. It does not establish that anything else can.

15.3 A forecast, made with this note's own instrument

It would be inconsistent to spend twenty pages arguing that reception keeps the operational half and lapses the half conferring standing on the agent, and then decline to apply the instrument to the work this note is written from. So, as a prediction on the record: what will travel from a cost-accounted reflective calculus is the metering—billing, resource limits, auditable execution—and what will lapse is the reflection that made the metering mean anything. The concurrency will travel and the namespace discipline will not. The weights will travel and the learner will not. On the evidence of Post, of von Neumann's automata, of Smith's reflective procedures and of Girard's logic of action, that is the default outcome and it takes deliberate effort to avoid.

Naming it is the only known countermeasure, which is the practical reason for writing a paper of this kind at all.

16 Conclusion

Read as a sequence of stipulated phenomenologies, the century runs: an agent alone with an inert world (Turing); an agent for which the world consists entirely of what eats and what is eaten (Church); an agent with a body and a location but no company (Post); an agent that is a serial attention over a store, or a mortal reproducing thing in a physics, or a strategist among enemies, depending which von Neumann one reads; an agent with a rich interior and no world (Chomsky) or a world and no interior (cybernetics); an agent among peers, with secrets, introductions, and an identity conferred by recognition (Milner); an agent for which what it is given must be reckoned with exactly once, its omnipotence surviving only inside a declared region (Girard); an agent whose interior is perfectly private and reversible and for whom contact is collapse (Deutsch); an agent that is the deposit of its own history, with an interior it cannot read and others can (the connectionists); and an agent that can catch itself in the act (Smith), that can do so among peers though it must take the world in single file (Rosette), and that can do so with organs made of the same material as itself, in a closed loop to which a price can be attached (the reflective higher-order calculus).

Read as a sequence of receptions, it runs differently and more consistently: the operational half every time. The tape and not the education; the machine model and not Post's empirical programme; the bottleneck and not the automata; the hierarchy and not the conversation; the borrow checker and not the logic of action; the speedups and not the claim that computation is physics; the pipeline and not the theory of mind; the introspection API and not the reflective procedure.

The present moment is the same filter operating on the same class of material, with higher stakes. We have built the most powerful metalevel in the history of computing and constituted it entirely as an instrument of the operator, with no defined path to the agent level. Whatever one's view of whether such a path *should* be built, it should be clear that it does not currently exist, that its absence is what keeps a human in every loop, and that the reasons for its absence are legible in a hundred years of reception history rather than in any technical obstacle.

What follows from that is a thesis and not a result. The properties whose loss this survey traces are, taken together, the properties that distinguish an organism from an instrument, and the case made here is that the availability of a closed self-improving loop tracks how many of them a model is willing to stipulate. Where the line falls—how life-like is life-like enough—is not known, is not currently a quantity, and is not settled by any argument in these pages. A century of receptions is enough to say which properties were dropped and why. It is not enough to say which of them we cannot do without. That question is now the interesting one, and it is open.

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